

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0109

Talenergy Martins Creek Llc Harwood 138 kV
December 04, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0109. The Interconnection Customer proposes a 228.3 MW natural gas combined cycle facility interconnecting at the Talenergy Martins Creek Llc Harwood 138 kV substation in the Pennsylvania area. The request was received on February 25, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 79 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$17,753,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Valley Home Tap 2-Newcomerstown 138 KV	West Gallain-Walnut 345 KV	130.2	228	7.3
E Arlington-Valley Home Tap 2 138 KV	Walut-W Gallatin 345 KV	121.8	258	7.3
Wescosville-East Arlington 138 KV	West Gallatin-Walnut Substation 345 KV L	111.1	163	5.8
Newcomerstown-Morovis 8801 Substation 138 KV	Walnut-West Galatin Substation 345 KV	108.0	233	7.3
Talenergy Martins Crk Llc Harwood-Paul 138 KV	Felida-Talenergy Martins Crk Llc Harwood 345 KV	107.3	279	39.4
Paul-Osha Star Worksite 138 KV	Talenergy Martins Crk Llc Harwood-Felida 345 KV	106.9	268	39.4
Wilson Dam-Tyrone Phelps Dodge 138 KV	Wigwam-Upsco 345 KV	105.2	280	5.4
Osha Star Worksite-Satsop 138 KV	Felida-Talenergy Martins Crk Llc Harwood 345 KV L	102.1	246	39.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Valley Home Tap 2-Newcomerstown 138 KV	Rebuild line with higher-capacity conductor	\$4,440,000	78% (\$3,468,000)
E Arlington-Valley Home Tap 2 138 KV	Rebuild line with higher-capacity conductor	\$2,503,000	78% (\$1,956,000)
E Arlington-Wescosville 138 KV Line	Rebuild line with higher-capacity conductor	\$1,632,000	100% (\$1,632,000)
Newcomerstown-Morovis 8801 Substation 138 KV	Reconductor line	\$1,485,000	78% (\$1,160,000)
Talenenergy Martins Crk Llc Harwood-Paul 138 KV	Reconductor line	\$1,282,000	100% (\$1,282,000)
Osha Star Worksite-Paul 138 KV	Reconductor line	\$2,261,000	100% (\$2,261,000)
Wilson Dam-Tyrone Phelps Dodge 138 KV line	Reconductor line	\$5,431,000	100% (\$5,431,000)
Osha Star Worksite-Satsop 138 KV	Reconductor line	\$563,000	100% (\$563,000)
	Total allocated to project		\$17,753,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.